Cancer Detective

Teacher notes

This resource invites students to take on the role of a research team to understand how cancer mutations are studied and how this knowledge can help choose the most appropriate treatment for each patient. Through a simulation activity inspired by real research from the BBGLab group at IRB Barcelona, students will discover what cancer driver mutations are and how personalized medicine can improve the treatment of this disease.

This activity is based on the interactive resource Cancer Detective, a simulation to delve into how cancer works and discover personalized medicine. The content and the simulation within the interactive tool serve as the basis for working on the proposed topics.

Educational levels	4th of ESO, Baccalaureate (High School), with the possibility of adapting it for university students.
Duration	50-60 minutes

Main Learning Objective

Cancers are unique to each person. Each patient's tumor harbours a specific set of mutations that make it different. Therefore, it is essential to study them at a molecular level to identify these alterations and use this information to make clinical decisions, such as choosing the most appropriate treatment for each patient.

Introduction

Prior Knowledge (10-15 min)

Before sharing the interactive tool with the students, we can propose an activity to mobilize prior knowledge about cancer and its treatments. We divide the students into small groups (4-5 people). On a worksheet, we ask them to reach a consensus on the answers to the following questions:

- ➔ What is cancer?
- ➔ What types of cancer do you know?
- ➔ How is cancer treated? What problems can conventional treatments have (chemotherapy…)?
- ➔ Have you heard of any new therapy or treatment against cancer?

We share the answers of the different groups with the whole class, looking for common ground and listing additional information (types of cancer, new treatments…).

Another option would be to ask these same questions to the whole class orally.

What is Cancer? (10-15 min)

The first part of the interactive tool (What is cancer?) will serve as theoretical material to introduce how cancer works and treatments based on personalized medicine. In this way, students will understand the importance of analyzing each patient's mutations, identifying which mutations are cancer driver mutations, and using this information to determine the most appropriate treatments.

To introduce the theoretical part, we can explain to the students that they will use an online simulator where they will analyze two cancer samples from two patients. To do this, they will need to review basic concepts on cancer first.

How can we work on this?

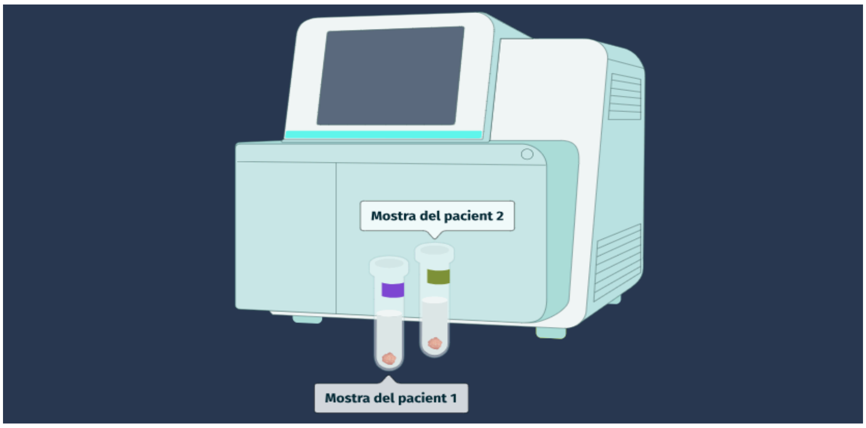
- ➔ You can use the materials from the interactive tool to explain this part to the students (if you prefer, you can adapt them into a presentation format).

- → You can propose autonomous work, individually or in pairs, and then share the main ideas extracted.

Simulation: We Are Cancer Detectives (30 min)

Are all cancers equal?

Next, we will explain that we are starting the research by simulating what happens when two tumor samples from oncology patients arrive at the hospital laboratory. With the help of the interactive tool, we will explain the four steps we will follow to carry it out.



We divide the class into two groups: one group will study skin cancer and the other will study lung cancer. Alternatively, we can have the whole class study the same type of cancer (e.g., skin) and then repeat the analysis with the other type (lung).

Once the type of cancer is chosen, the students, working in pairs, will study the clinical characteristics of the two patients and answer the question: **Do you think the cancers of the two patients will have the same mutations?**

We record the students' answers in a joint table and ask them to explain the reason for their choice.

Do you think the cancers of the two patients will have the same mutations?	**Yes**	**No**
Number of pairs that choose this option **before** sequencing the DNA of the samples		
Number of pairs that choose this option **after** sequencing the DNA of the samples		

To check their answer, they will analyze the mutations of the two patients and compare them between groups. Here we can introduce the final objective of the simulation: identifying these mutations will also help them find the most appropriate treatment for the patients.

We clarify that each pair has received two unique and different samples (within the simulation) and that, once they have finished the analysis, we will compare the results obtained by each of the pairs.

Research: Sequencing and Analysis

We instruct students to perform the sequencing of the DNA samples of the two patients using the simulator. Next, we ask them to analyze the results they have obtained and to answer the question "Do you think the cancers of the two patients will have the same mutations?" from the previous table again.

We explain that in this first analysis we have obtained the list of mutated genes for each patient, as well as the change that occurs in the corresponding protein

Once they have recorded all the answers in the table, **we will have discovered that the mutations of the two patients will always be different.**

We can share the results and propose the following questions:

- ➔ Who has a sample with mutations in the BRAF gene (in the case of skin cancer)? And in EGFR (in the case of lung cancer)?
- ➔ Of those who have a sample with altered BRAF, what mutations or changes in the protein are observed?

Additionally, we can also ask:

- ➔ If we compare the mutated genes among patients with the same type of cancer, we can look for examples of genes and mutations within them that are recurrent and appear in several patients.

Research: Driver Mutations and Treatments

We explain that, once the mutations are identified, we need to find out which ones might be cancer driver mutations. Students will write down on their worksheet which driver mutations they have obtained. In the group discussion, we can create a joint class list and note whether there are mutations that are repeated more or less frequently.

Next, students will discover for which driver mutations a targeted therapy exists. We write down the treatment, if there is one, for the driver mutations on the board.

Evaluation

Once the core activity is finished, students can individually or in pairs answer 10 questions to evaluate their knowledge.

Conclusions

Once all the results have been collected, we ask the students to write down on their worksheet the conclusions they have reached with their research.

Next, they will share their conclusions with another pair and reach a consensus. At this point, we will ask each group of 4 to reflect on the following topics:

- ➔ Why is it necessary to study the mutations of each cancer?
- ➔ Why is cancer research important?

Finally, we share the results of all the groups and conclude by explaining that, as a conclusion of the research, we can say that:

Once the samples are analyzed, we can see that the result of our research demonstrates that each patient's cancer is different, and that, in each case, it will be necessary to identify the mutations present, analyze which ones are driver mutations, and establish whether targeted treatments are available for them.

Additional Work

Downloading the Results

Students can download the simulation data as a table. This data can be compared with the results of a second round of simulations with the type of cancer that has not been analyzed yet.

Additionally, the results of the whole class could also be aggregated in a shared spreadsheet to analyze different topics, such as which mutation is more frequent or for what percentage of mutations we have a treatment available.

Supplementary Material

In this section, you will find the basic concepts of cancer genomics research and precision oncology. You can recommend that students consult it after the theoretical introduction and at the end of the activity. Once they have done so, we can jointly review the most advanced concepts (Cancer Genomics, Bioinformatics, and BBGLab Bioinformatics Tools).

Cancer Detective

Student Worksheet

Prior Knowledge

In small groups or pairs, answer these questions together:

What is cancer?

What types of cancer do you know?

How is cancer treated? What problems can conventional treatments have (chemotherapy…)?

Do you know of any new therapy or treatment to treat cancer?

Share your answers with the class and complete the list of cancer types and treatments.

What is Cancer?

Read or listen carefully to the explanation about how cancer works and personalized medicine. Take notes if you need to.

Simulation: We Are Cancer Detectives

1. Work in pairs.
2. Divide the pairs by cancer types:
   - ➔ Group 1: Skin cancer
   - ➔ Group 2: Lung cancer
3. Using the interactive tool, analyze the two patient samples.

Before sequencing the samples, answer:

Do you think the cancers of the two patients will have the same mutations?	**Yes**	**No**
Answer		

Give reasons for your answer:

4. Carry out the sequencing of the samples using the interactive tool.
5. Analyze the mutated genes and the changes in the proteins.
6. Note down whether they match or not.

Answer again:

Do you think the cancers of the two patients will have the same mutations?	**Yes**	**No**
Answer		

Give reasons for your answer:

Note down which driver mutations you have found and if there is any treatment available:

Driver mutation	Available treatment

Conclusions

Fill in this section once you have completed the evaluation.

➔ Summarize what you have discovered in your research.

➔ Share it with another pair and write a **joint conclusion.**

Once you have completed the evaluation section, answer the following questions, according to the type of cancer you have studied:

Whose sample would you expect to have more age-related mutations?

Skin cancer: What level of sun exposure do you think will cause the most mutations in the skin?

Skin cancer: What level of sun protection do you think will best reduce the risk of mutations in the skin?

Lung cancer: What type of tobacco consumption do you think will increase the risk of accumulating mutations in the lungs the most?

Lung cancer: What type of exposure to tobacco do you think will have the lowest risk of causing mutations in lung?

To finish, discuss the following questions within your group. Note down your answer and share it with the rest of the class.

Why is it necessary to study the mutations of each cancer?

Why is cancer research important?

Remember to check the "Want to know more?" section to delve deeper into concepts like cancer genomics and bioinformatics!